

EFFECT OF ANTITHYROID AGENT (CARBIMAZOLE) ON THYROID GLAND OF GUINEA PIG WITH REFERENCE TO HYPERTROPHY OF FOLLICULAR CELLS

***TASNEEM FATIMA AND M. YUNAS KHAN**

*Anatomy Department, Basic Medical Sciences Institute,
Jinnah Post-graduate Medical Centre, Karachi Pakistan.*

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ABSTRACT

Effect of Carbimazole (antithyroid drug) on thyroid gland of guinea pig was noted. Normal healthy male guinea pigs were divided into two groups, one group was given carbimazole treatment orally for 30-days, the other group remained untreated and served as control.

Histological observations showed enlargement (hypertrophy) of thyroid follicular cells with a significant decrease in the diameter of lumina of follicles, amount of colloid and some increase in the diameter of follicles measured.

An increase in connective tissue elements and vascularity was noted too. But all these changes were not very extensive because the duration of goitrogen (Carbimazole) treatment was too short (30 days) for such an animal (guinea pig) whose thyroid hormone metabolism is very slow with delayed TSH response to alter changes in histological picture of thyroid gland.

INTRODUCTION

A wide variety of chemical agents have the ability to inhibit one or more reactions required in the syntheses of thyroid hormones. When the effect of such agents is sufficient to reduce the secretion of thyroid hormones to subnormal levels, secretion of TSH is increased and goiter ensues. Hence such agents are commonly termed goitrogens. In clinical practice, goitrogenic agents are used as drugs for the treatment of hyperthyroidism (William's, 1974). There are many goitrogenic agents including thionamid group.

Like thiouracil and propylthiouracil, Carbimazole is one of the thionamide group of goitrogen. It is derived from methimazole (Lawson and Barry 1951) by replacement of the hydrogen atom of Sulfahydral group of methimazole by a Carbethoxy radical. The action of Carbimazole was found to be slightly more potent than methimazole in inhibiting thyroidal uptake of I^{131} in both rat and man (Lawson and Barry 1951). The advantage of this derivative (*carbimazole*) over the parent substance (methimazole) was its purportedly diminished catabolism with slow hydrolysis of the carbethoxy group in body to change into methimazole, in this manner a steady supply of the inhibiting compound would be available to the thyroid gland. In turn smaller doses would be required and thus appreciably reduce the incidence of toxic reactions (Selenkow and Callaco, 1961). The

**Present address:*

Tasneem Fatima

Instructor III Anatomy Department.

Aga Khan Medical University & Hospital Karachi - Pakistan.

M. Yunas Khan

House 785, Sector J-91, Islamabad.

carbimazole depresses the formation of thyroid hormone by partly reducing the uptake and concentration of inorganic iodine by the thyroid gland (Martindle, 1972). also by blocking the organification of iodine by suppressing the activity of enzyme peroxidase, which is responsible for the oxidation of iodide (Taurog *et al.*, 1970; Taurog, 1976 cited by Goodman and Gilman, 1995). (3) by inhibiting the coupling of two iodinated tyrosines to form thyroxine (T₄) or tri-iodothyronine (T₃) (guyton, 1996). Hence by suppressing the formation of thyroid hormone it indirectly increases the output of TSH from adeno-hypophysis (Redmond and Tuffery, 1981) induces thyroid gland to undergo hypertrophy and hyperplasia.

The effect of carbimazole was found to be very quick and effective in case of rat to produce a three fold increase in hypertrophy and hyperplasia of thyroid follicular cells after three weeks treatment (Redmond and Tuffery, 1979). Guinea pig was found to be quite resistant to other goitrogens like thiouracil and propylthiouracil to become hypothyroid due to its slow rate of peripheral utilization of thyroid hormone which cause a delay in TSH secretion from anterior pituitary so the thyroid gland remains unstimulated i.e in a euthyroid state for a long period. Therefore a prolonged treatment of these goitrogens was required to render the animal hypothyroid. Keeping in view the efficiency of carbimazole in producing goiter in rat it was decided to test the effect of this particular drug in guinea pigs model of animal.

MATERIALS AND METHODS

Male guinea pigs of J.P.M.C breed weighing between 200 and 300 grams were used for this study. All the animals were kept in an ideal laboratory conditions and were supplied with excess of food and water adlibitum. These experimental animals were divided into two groups. One group of animals given carbimazole. Orally (0.13 mg of drug in 0.1 ml of solution per 300 gm body weight) for thirty days and this group of animal is regarded as "treated group". The other group of animals remained untreated and served as control.

On the last day of treatment all the animals including controls were killed by over dose of ether anaesthesia, the thyroid glands were removed, fixed in Bouin's fixative, dehydrated, infiltrated and embedded in paraffin, five micron thick sections were cut and stained with Haematoxylin & Eosin stain (Bancroft & Harry, 1994).

OBSERVATION AND RESULTS

Observation were based on :

Microscopic observations:

Degree of hypertrophy of follicular cells by measuring.

- i) Diameter of the follicular lumina.
- ii) The outer diameter of follicles.

Approximately three hundred follicles (situated in the center of the lobes of thyroid gland) were studied from each animal. The diameter of follicles was measured with an ocular micrometer which was calibrated against a stage micrometer.

i) The diameter of follicular lumina:

Control animals: The mean diameter was $63.52 \pm 3.66 \mu\text{m}$.

Treated animals: The mean diameter was $49.08 \pm 4.44 \mu\text{m}$.

This value in treated animals was much less than control, and the difference was statistically significant ($P < 0.05$, Table 1).

ii) The outer diameter of follicles:

Control animals: The mean diameter of follicles recorded was $72.25 \pm 7.95 \mu\text{m}$.

Treated animals: The mean diameter of follicles measured as $77.45 \pm 6.24 \mu\text{m}$, though the mean outer diameter of follicles in control glands was less as compared with follicles of treated glands but the difference was statistically not significant ($P > 0.1$, Table 2).

Thus there is apparently an increase in follicular diameter with decrease in lumen of treated animals due to increase cell height.

Microscopic Observation:

Under light microscope the gland observed is covered by fibrous capsule from which fibrous septa pass into the substance of gland but do not divide, the gland clearly into lobes and lobules (Figs. 1 & 2).

The blood vessels run through these septa into the gland. The blood supply of the thyroid gland is rich, extensive capillary network were observed surrounding the follicles. The gland is consist of follicles and interfollicular stroma. Follicles differ in size and are rounded, oval or oblong in shape. The peripheral follicles are larger in size than the centrally placed ones.

Control Animals:

The epithelial or follicular cells are squamous in shape and arranged in a single layer enclosing a large central cavity or follicular lumen filled with homogenous colloid with numerous small and large vacuoles within the colloid nearer the apical ends of the follicular cells (Fig. 3). In the central follicles the colloid has numerous vacuoles and stained light pink with Haematoxylin & Eosin. Where as the colloid in peripheral follicles has less vacuoles and stained slightly deeper pink.

Between the follicles, connective tissue stroma consists of connective tissue fibers and cells more commonly the fibroblasts was found present within the stroma rich in blood capillary network.

Treated group: after 30 days of treatment with carbimazole. The histological changes observed were an increase in the height of individual follicular cells from squamous to high cuboidal (Fig. 4). Mitotic figures were observed in follicular and stromal cells population. The lumina of the follicles decreased in diameter but did not obliterate, Slight reduction in the amount of colloid was observed. Blood vessels were enlarged, the amount of connective tissue stroma become slightly dense.

DISCUSSION

The results found in the present study of thyroid gland in guinea pig treated with carbimazole for 30 days reflected the hypertrophy i.e an increase in size and height of the thyroid follicular cells which has been demonstrated by measuring diameter of follicles and the diameter of their lumina in control and treated animals. Besides this an increase in connective tissue element between the follicles with increased vascularity was also observed.

Histological changes which take place in the thyroid gland after treatment with antithyroid substances, which blocks the formations of thyroid hormones have been well documented in literature.

An increase in the individual volume of the thyroid follicular cells, which bulges into, and obliterates most of the follicular lumina with very little stored colloid, increased vascularity in thyroid gland of rat after 30 days of thiouracil treatment (Santler, 1957). Increase in the height of the follicular cells, reduction in the amount of colloid and increase in vascularity in rat by the end of the 3rd week of carbimazole treatment (Redmond and Tuffery, 1979). The hypertrophy of the follicular cells led to the almost complete obliteration of the lumina so that the normal structure of the tissue was obscured after a 10 to 12 weeks carbimazole treatment. (Redmond and Tuffery, 1978). Longer exposure to carbimazole in rats resulted in the apparent obliteration of the follicular lumina caused by the hypertrophy of the follicular cells and almost total absence of colloid. After 4-6 months of treatment, showed localized papillary changes and occasional foci of pyknotic or fragmented nuclei (Redmond and Tuffery, 1981). Similar changes have been described in rats fed thiouracil for 6 months or more (Wollman, 1961).

The maximal histological changes in thyroid gland of guinea pig were achieved after 100-200 days of prolonged continuous propyl thiouracil treatment (D'Angelo *et al.*, 1954). This delayed goitrogenesis in guinea pig was proposed due to a relatively slow rate of peripheral utilization of circulating thyroid hormone with consequent lag in pituitary TSH response (Rand *et al.*, 1951). The response of the pituitary thyroid system to antithyroid drugs is five times more in rat than guinea pig therefore a short term goitrogen therapy in rat is sufficient to stimulate and increase the pituitary TSH secretion which stimulates the thyroid gland, causes loss of colloid, increased vascularity and hypertrophy and hyperplasia of cells.

A rise in TSH concentration in guinea pig adenohypophysis could only be possible after a prolonged treatment. (D'Angelo, 1953) demonstrated a rise in TSH concentration in guinea pig after 100-200 days continuous propylthiouracil treatment. But present study the period of carbimazole treatment was too short i.e 30 days to effect maximum hypertrophy and hyperplasia in thyroid follicular cells as has been observed in rat after 3 weeks carbimazole treatment (Redmond and Tuffery, 1979), this might be due to mild TSH response because of relatively slow peripheral circulation of thyroid hormone in guinea pig which gives a picture of thyroid gland not much active histologically.



Fig. 1. Section of thyroid gland showing connective tissue capsule (C) surrounding the gland.
H & E stain, photomicrograph, approx. 40 X.

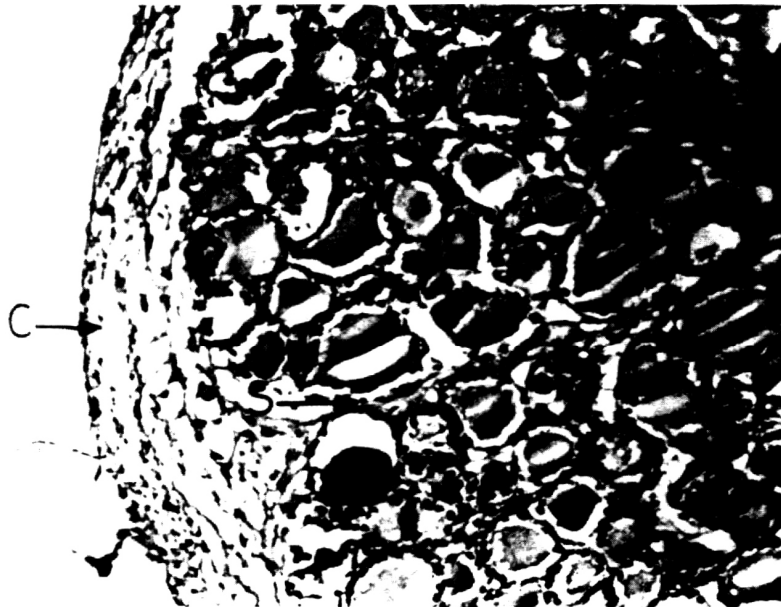


Fig. 2. Section of thyroid gland of guinea pig showing connective tissue septa (S) traversing within the glandular substance.
H & E stain, photomicrograph, approx. 100 X.

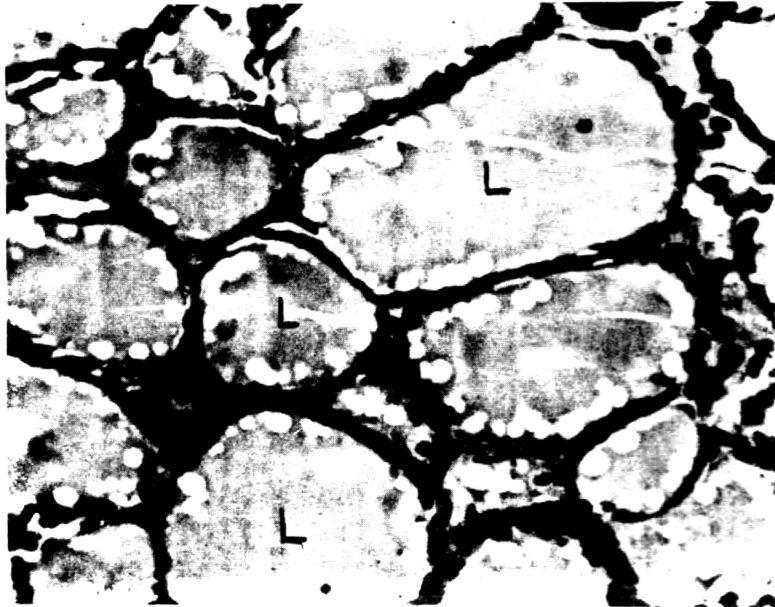


Fig. 3. Section of thyroid gland of guinea pig (control) showing squamous type of follicular epithelium surrounding large lumens (L).
H & E stain, photomicrograph, approx, 650 X.



Fig. 4. Section of thyroid gland of guinea pig (treated with Carbimazole) showing high cuboidal type of follicular epithelium surrounding small lumens (L).
H & E stain, photomicrograph, approx, 650 X.

TABLE -1
Mean Diameter of the Follicular Lumina.

Animal-Group	No.of Animals	Mean Diameter of the Follicular Lumina (μm)	Statistical Significance
Control	6	63.52 $\pm$ 3.66	
Treated	6	49.08 $\pm$ 4.44	P<0.05 Significant

TABLE - 2
Mean Outer Diameter of the Follicles.

Animal-Group	No.of Animals	Mean Outer Diameter of the Follicles (μm)	Statistical Significance
Control	6	72.25 $\pm$ 7.95	
Treated	6	77.45 $\pm$ 6.24	P>0.1 Not Significant

REFERENCES

- Bancroft, J.D. & Harry, C.C. Manual of Histological techniques and their Diagnostic Application, Edinburgh, Churchill livingstone 1994.
- D'ANGELO, S.A., (1953). Cytophysiologic aspects of thyrotrophic hormone secretion in the goitrous guinea pig. *Endocrinology*. **52-53** : 331-337.
- D'Angelo, S.A., Stevens, C.E., Paschkis, K.E., Cantarow, A., Sunderman, W. & Friedler, G. (1954). The effect of goitrogen withdrawal on the Pituitary thyroid system of guinea pig. *Endocrinology*. **54** : 565-579.
- Fawcett, D.W. (1993). A Text book of Histology, 12th ed. Champman & Hall, New York, N.Y.
- Goodman, L.S. & Gilman, A. The Pharmacological basis of Therapeutics. 9th ed. New York. Macmillan. 1995.
- Guyton, A.C. Textbook of Medical Physiology, 9th ed. Philadelphia. Saunders. 1996.
- Lawson, A. & Barry, G. (1951). Treatment of thyrotoxicosis with 2-Carbethoxythio-1-methyl-glyoxylin (Carbimazole) *Lancet*. **2** : 621.
- Martindle, W. Extrapharmacopiea, 26th ed. London, Pharmaceutical Press. 1972.
- Rand, C.C., Lee, N.D. & Williams, R.H. (1951). Factors contributing to the difference in

- propylthiouracil goitrogenesis in rats and guinea pig. *Proceeding Society Experimental Biology Medicine*. 78 : 307-309.
- Redmond, O. & Tuffery, A.R. (1978a). Mitotic rate of the rat's thyroid gland during hypertrophy induced by an antithyroid agent (carbimazole). *Journal of Anatomy*. 127 : 353-362.
- Redmond, O. & Tuffery, A.R. (1979). Mitotic rate of thyroid follicular cells in untreated and goitrogen treated rats: variation with time of day *Journal of Anatomy*. 129 : 731-741.
- Redmond, O. & Tuffery, A.R. (1981). Thyroid proliferation, body weight, thyrotropin and thyroid hormones in chronic antithyroid (carbimazole) treatment in rats. *Journal of Anatomy*. 133 : 37-47.
- Santler, J.E. (1957a). Effect of cessation of thiouracil administration on the cell population of the thyroid of rats. *Journal of Endocrinology*. 15 : 1-8.
- Santler, J.E. (1957b) Growth in the cell populations of the thyroid gland of rat treated with thiouracil. *Journal of Endocrinology*. 15 : 151-161.
- Selekow, H.A. & Callaco, F.M. (1961). Clinical Pharmacology of antithyroid compounds. *Clinical Pharmacology & Therapeutics*. 2 : 191-219.
- Taugog, A., Lothrop, M.L. & Estabrook, R.W. (1970). Improvements in the isolation procedure for thyroid peroxidase : nature of heme prosthetic group. *Biochemistry Biophysics*. 139 : 221-229.
- Williams, R.H. Textbook of Endocrinology, 5th ed. Philadelphia. Saunders. 1974.
- Wollman, S.H. & Breitman, R.R. (1970). Changes in DNA & weight of thyroid gland during hyperplasia and involution. *Endocrinology*, 86 : 322-327.